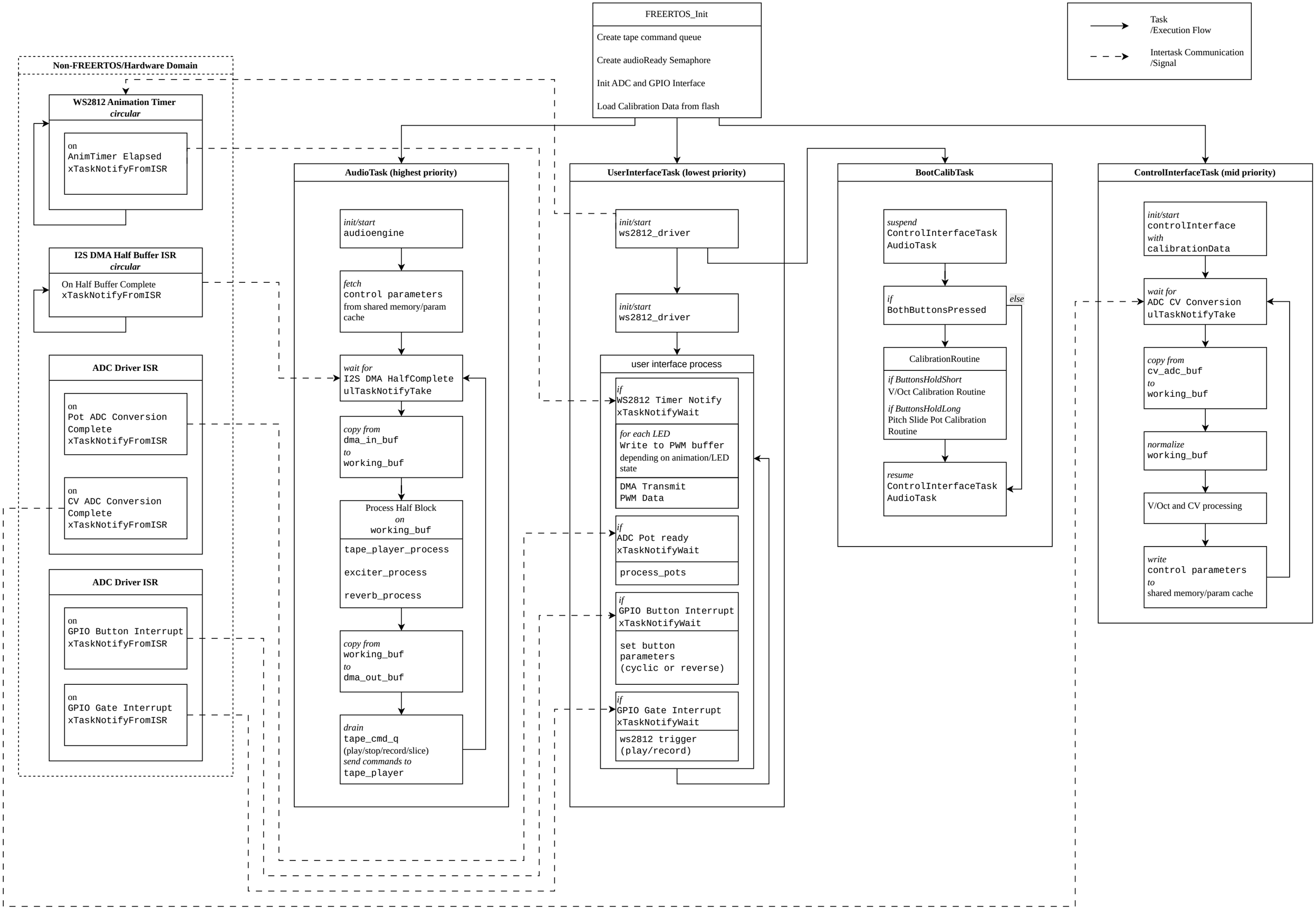
FREERTOS_Init
Create tape command queue
Create audioReady Semaphore
Init ADC and GPIO Interface
Load Calibration Data from flash
Task /Execution Flow
Intertask Communication /Signal
Non-FREERTOS/Hardware Domain
WS2812 Animation Timer
circular
on AnimTimer Elapsed xTaskNotifyFromISR
I2S DMA Half Buffer ISR
circular
On Half Buffer Complete xTaskNotifyFromISR
ADC Driver ISR
on Pot ADC Conversion Complete xTaskNotifyFromISR
on CV ADC Conversion Complete xTaskNotifyFromISR
ADC Driver ISR
on GPIO Button Interrupt xTaskNotifyFromISR
on GPIO Gate Interrupt xTaskNotifyFromISR
AudioTask (highest priority)
init/start audioengine
fetch control parameters from shared memory/param cache
wait for I2S DMA HalfComplete ulTaskNotifyTake
copy from dma_in_buf to working_buf
Process Half Block on working_buf
tape_player_process
exciter_process
reverb_process
copy from working_buf to dma_out_buf
drain tape_cmd_q (play/stop/record/slice) send commands to tape_player
UserInterfaceTask (lowest priority)
init/start ws2812_driver
init/start ws2812_driver
user interface process
if WS2812 Timer Notify xTaskNotifyWait
for each LED Write to PWM buffer depending on animation/LED state
DMA Transmit PWM Data
if ADC Pot ready xTaskNotifyWait
process_pots
if GPIO Button Interrupt xTaskNotifyWait
set button parameters (cyclic or reverse)
if GPIO Gate Interrupt xTaskNotifyWait
ws2812 trigger (play/record)
BootCalibTask
suspend ControlInterfaceTask AudioTask
if BothButtonsPressed
else
CalibrationRoutine
if ButtonsHoldShort V/Oct Calibration Routine
if ButtonsHoldLong Pitch Slide Pot Calibration Routine
resume ControlInterfaceTask AudioTask
ControlInterfaceTask (mid priority)
init/start controlInterface with calibrationData
wait for ADC CV Conversion ulTaskNotifyTake
copy from cv_adc_buf to working_buf
normalize working_buf
V/Oct and CV processing
write control parameters to shared memory/param cache